

Jeng Yang Kong (Jayden)

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SUMMARY

Software Engineer with 4 years of experience and a Master's in Data Science (CGPA: 4.0), focused on production software and applied AI — specialising in multi-agent LLM systems, RAG pipelines, agentic workflows, LangGraph orchestration, LLM evaluation, and domain-specific fine-tuning.

TECHNICAL SKILLS

AI & ML: LangGraph, RAG, Hybrid Retrieval, Reranking, LLM Evaluation, Prompt & Context Engineering, Agentic Workflows, Function Calling, LangChain, LangSmith, LlamaIndex, pgvector, SFT, QLoRA, Optuna,
Backend & Data: Python, Flask, FastAPI, PostgreSQL, Supabase, REST APIs, Docker, CI/CD, Pandas, SQL
Frontend & Mobile: TypeScript, React, Next.js, Vue.js, Flutter, Swift
Cloud & Tools: Gemini API, OpenAI API, Claude API, AWS, Vercel, Git

PROFESSIONAL EXPERIENCE

Software Engineer

Jul 2023 – Present

Daikin Singapore

Singapore

- Architected and shipped a RAG support chatbot with Gemini embeddings, page-aware chunking, hybrid BM25 and dense retrieval, per-product filtering, SSE streaming, and citation chips linking to source PDF pages; added RAGAS evaluation pipeline achieving 0.88 faithfulness and 0.87 answer relevance.
- Deployed SSD MobileNet occupancy detection into room monitoring app, integrating ML inference with HVAC control logic to automate air-conditioning decisions in commercial buildings.
- Built a database backup tool for Reiri HVAC edge controllers, adding automatic disconnection detection and record backfilling to preserve SQLite data integrity under unstable networks.
- Rebuilt the Reiri Office frontend in Vue.js for the Reiri Touch smart controller, delivering a responsive, touch-optimized interface for centralized control of air conditioning, lighting, and connected devices.
- Engineered a real-time energy analytics dashboard for building managers and optimized historical HVAC data retrieval by tuning batch size, sequencing, and recovery flows, reducing first-load time by 25%.
- Built internal QC tools for SQLite-based HVAC testing, including realistic operational dataset generation and a side-by-side database comparison tool, reducing QA data preparation time by 80% and improving issue validation.

iOS Engineer

Jul 2021 – Jul 2022

FPLE Group

Kuala Lumpur, Malaysia

- Built and shipped a production eCommerce iOS app in Swift (MVVM, Combine, UIKit), covering secure auth, product browsing, cart, and order processing with REST API integration.
- Designed reusable UIKit components following Apple HIG, reducing UI code duplication by 30% and ensuring cross-device consistency.
- Applied SOLID principles to maintain a modular, scalable codebase; performed root-cause analysis and defect resolution to improve delivery reliability.

PROJECTS

PolicyLens: Multi-Agent AI Policy & Regulation Research System

Apr 2026 – Jul 2026

[Project Page](#) | [Source Code](#) | [Side Project](#) | *LangGraph, FastAPI, PostgreSQL, Next.js*

- Built a full-stack AI policy research system with LangGraph, FastAPI, PostgreSQL, and Next.js, supporting end-to-end workflows from ambiguous query to streamed, citation-backed report and follow-up chat.
- Designed a multi-agent orchestration engine with 4+ specialized LLM roles and parallel Send fan-out, enabling concurrent subagents to search, fetch, extract, and synthesize structured findings.
- Engineered a 3-layer memory architecture using typed graph state, context compaction, and PostgreSQL checkpointing, enabling resumable runs, persisted follow-up chat, and human-in-the-loop clarification.
- Improved citation faithfulness from 72% to 89% across 20 benchmark queries by adding schema-validated evidence spans, targeted gap research, and automated grounding checks.

- Built an agentic personal finance assistant using function calling, multi-turn dialogue management, structured outputs, dynamic intent routing, and chart generation tools that translate user queries into visual spending.
- Implemented a RAG pipeline for financial document intelligence with chunking, embedding retrieval, prompt templates, and context injection to improve relevance and reduce hallucinations.
- Integrated Gemini Vision for receipt OCR, extracting structured transaction data from uploaded images and injecting parsed results into the financial assistant’s context for downstream query and analysis.
- Developed a multi-user backend with Supabase row-level security, role-based access control, and CI/CD deployment on Vercel; refined retrieval quality based on real usage patterns.

- Fine-tuned LLMs using SFT and QLoRA for financial-domain sentiment classification and content understanding, improving task performance on benchmark evaluation (99.3% test accuracy).
- Collected and processed 160k+ financial news articles via Polygon.io API, curating 35k high-quality labeled samples and engineering features to capture temporal market signals.
- Applied hyperparameter optimization across LSTM, XGBoost, and Random Forest models on OHLCV and sentiment sequences, reducing TSLA next-day price prediction RMSE by 10.9% and improving R^2 by 4.4%.
- Investigated hallucination reduction and domain adaptation of LLMs in financial NLP, building reproducible evaluation benchmarks across model families.

EDUCATION

Universiti Malaya <i>Master of Data Science (CGPA: 4.0)</i>	Jul 2023 – Aug 2025
Monash University <i>Bachelor of Computer Science (Advanced)</i>	Jul 2018 – Jul 2021